SCFT-A: Editorial Assessment and Publication Plan

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Recommendation. Present this work as a restricted classical consistency construction. My first journal recommendation by subject fit is *General Relativity and Gravitation* (GRG), after aligning the title and abstract with that restricted contribution and resolving its submission requirements. Its scope includes alternative gravity and mathematical relativity. PRD is also within scope, but its emphasis on significant advances makes clear novelty and physical scope especially important. No verified acceptance-rate evidence supports calling either journal the easiest, and neither acceptance nor a numerical quality rating can be guaranteed. See the official GRG scope and PRD scope.

Submission status. The supplied material supports a positive scalar quadratic result and a reproducible finite-time response. It does not establish a completed observational replacement for dark matter and dark energy. The revised manuscript is the strongest constrained revision prepared here; it should not be represented as independently peer-reviewed or guaranteed publishable. The fixed-coefficient stationary approximation has a substantive scale problem described below.

What the revision preserves and changes

The abstract and the source through the end of page five are protected. Their scientific wording, equations, citation order, and forward-reference values are retained. The uploaded PDF calls the endpoint result Theorem XII.1, while the uploaded source used a proposition environment. The environment outside the protected pages has been corrected to reproduce the PDF's designation consistently.

The main manuscript retains the completed action, the exact constraint and scalar calculations, the local-transition proof, the de Sitter polynomial certificate, and the reproducible matter response. Expanded cubic calculations and original Appendices A–M have been placed in the technical companion. The discussion and conclusion now distinguish mathematical consistency from observational viability. The reference action is defined consistently, the disjoint DBI and late branches are distinguished, and the actual high-charge density and galactic scale limitation are stated explicitly. A version-specific GitHub data statement appears at the end of the manuscript.

Keeping the protected opening's many forward references fixed constrains how aggressively the preceding equation sequence can be shortened. PRD research articles have no fixed length limit; excessive breadth and unsupported implications matter more than page count alone. The historical reference was used only to consider the progression from definitions to derivation to consequences. No text, equations, or scientific claims from that paper have been inserted. Modern journal practice also requires clear prior-work comparison, reproducibility, and explicit limits. See PRD author guidance.

Journal-specific constraint

GRG currently requests a 150–250-word abstract and four to six keywords, and states that the abstract should make the new result clear. The protected abstract exceeds that range, and the protected keywords exceed that count. The preservation instruction therefore prevents a fully compliant GRG submission package. A shorter abstract must be explicitly authorized before it is substituted. The broad opening also needs to match the restricted result; merely shortening it would not settle the scientific issue. GRG's required declarations must be completed truthfully by the author. See GRG submission guidelines.

Scientific findings that affect publication

1. The central positive result survived the checks. The public generator's 16 symbolic checks passed when rerun without changing its source. The full 42-mode scan, its refinements, and the three RK45 comparisons were rerun. In this environment the largest sampled response reproduced the manuscript's stored binary64 value

$$1.0463168638938594.$$

This decimal is a numerical output, not an exact real-valued bound. The endpoint sign is instead established by its symbolic positive-coefficient polynomial. Eight additional exact checks verified the local polynomial with both switch derivatives, reciprocal endpoint identities, spatial-scale identities, and the ordering of the two clock intervals. Passing these implemented checks does not establish all claims of the theory.

2. The fixed benchmark does not establish its galactic MOND approximation. Its coefficients give the exact identities

$$\kappa_m = 120H_d^2, \qquad \ell_0 = \kappa_m^{-1}, \qquad \eta_R = (2\kappa_m)^{-1},$$
$$\frac{\ell_0 + 8\eta_R}{L^2} = \frac{1}{24(H_d L)^2}, \qquad \frac{2\ell_0 + 8\eta_R}{L^2} = \frac{1}{20(H_d L)^2}.$$

The manuscript's controlled AQUAL reduction requires these corrections to be small. They grow without bound as $H_d L \to 0$. Thus the fixed coefficients cannot justify that reduction in a galactic subhorizon limit. This calculation does not exclude every nonlinear sourced solution; it rules out the particular small-correction argument. The positive scalar theorem does not resolve this physical limitation.

3. The low-charge matter-like interval is a different branch regime. The exact ordering is

$$Q_b = \frac{101}{100} < Q_L = \frac{203}{200} < Q_d = \frac{103}{100}.$$

The displayed cosmological trajectory approaches Q_d from above. It does not visit the lower DBI interval on that approach. Its early behavior comes from the high-charge exponential continuation. On that continuation, with $F = C_T a^{-3}$,

$$\frac{\rho_T}{M_{\rm Pl}^2} = \left(Q - \frac{1}{8}\right) C_T a^{-3} - K_E + \frac{F_E}{8}.$$

Because Q changes logarithmically with a, $a^3\rho_T$ is not constant. A small asymptotic pressure fraction is insufficient to establish realistic finite-redshift clustering.

4. The remaining positive results have defined limits. The lapse and trace inverses require their stated domains and coercivity assumptions. General nonlinear evolution and its loss-free energy estimate remain unproved. Perfect-fluid responses omit the microphysics needed for a primordial Boltzmann calculation. No interaction cutoff, galaxy matching solution, or observational likelihood is provided. The parameter H_d is specified in the action; the construction does not predict its measured value or solve quantum vacuum-energy naturalness.

The numerical input for δ is an inherited, uncertified extrema evaluation. The endpoint proof covers a full exact interval and does not depend on its printed decimal precision. The independent numerical reproduction does not certify that the decimal equals the global-extremum definition of the action coefficient.

What belongs in the repository and what could become another paper

Material	Destination	Condition
Geometric identities, current, minisuperspace and constraint details	Technical companion and versioned repository	Supporting derivations of the present work, not a separate publication.
Reference-action reductions and historical diagnostics	Repository, labeled by action version	Retain for provenance; do not mix their modes with the completed action.
Source code, exact reports, input coefficients and response data	Versioned repository	Keep a fixed commit or archived release tied to the submitted manuscript.
Expanded interaction vertices	Companion now; possible later research	A later paper needs an actual canonical interaction-scale analysis, including matter and relevant higher-order terms.
Nonlinear evolution	Possible later mathematical paper	First prove a local solution theory, constraint propagation, and the required energy estimates.
Galaxies and lensing	Possible later phenomenology paper	First establish a compatible coefficient hierarchy and solve sourced equations with cosmological matching.
Primordial cosmology	Possible later observational paper	Implement the full perturbation and matter microphysics, initial statistics, and data comparisons.

These should become separate papers only if each supplies a new completed result. Dividing the present manuscript into several pieces does not by itself create publishable contributions. The technical companion should accompany this research as supporting material where the selected journal permits it.

Practical submission decision

For the present result, use the completed scalar consistency construction as the central contribution. Distinguish it explicitly from kinetic gravity braiding, relativistic khronon models, and established higher-spatial-derivative operators. Retain the scale limitation rather than implying that all desired dark-sector phenomenology has been shown.

Before submission to GRG, the protected title/abstract/keywords need an authorized journal-specific revision, and the author must complete the required declarations. For PRD, the present format is appropriate for a research article, but length flexibility does not address the scientific scope problem. No editorial correspondence was supplied, so the reasons for the earlier immediate rejections cannot be determined from this revision alone.

Reproducibility and file use

The checked public version is SCFT-A commit 2355864. Its full identifier and generator are included in the verification records. The package also contains the new independent-check script, its exact output, the rerun summaries, and the protected-page comparison. The LaTeX files embed their figures and use a manual bibliography. Compile each with `pdflatex` repeatedly until cross-references settle; a BibTeX run is unnecessary.

Substantive AI assistance is disclosed in the manuscript, as requested by the applicable journal policies. That disclosure does not assert that the author has already independently validated the revision. The scientific content and final submission decisions remain the author's responsibility. See the APS guidance on substantive AI assistance and the GRG author guidance.